Sharp convergence and phase selection for the primal-first Chambolle–Pock map at $\theta = 1/4$

Abstract

For the literal primal-first Chambolle–Pock iteration with extrapolation 1/4, we prove weak convergence for $\tau\sigma\|L\|^2 < 8/3$ and boundedness at equality. At the endpoint, every trajectory selects a single two-cycle: its even and odd subsequences converge weakly to the two states of an exact orbit, whose midpoint is a saddle point. Adjacent midpoints and ordinary Cesàro averages converge to that same saddle point. In finite dimensions all these convergences are strong. We characterize every endpoint two-cycle and prove that strict convexity of either function excludes nonconstant cycles. A four-pair storage certificate is given with complete rational coefficients and a reproducible exact-arithmetic verifier. Its alternating kernel explains both the sharp threshold and the endpoint behavior.

Keywords: primal–dual splitting; endpoint convergence; two-cycles; exact rational certificate

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1 The map and its sharp conclusions

Let $\mathcal{H}, \mathcal{G}$ be real Hilbert spaces, $f : \mathcal{H} \to (-\infty, +\infty]$ and $h : \mathcal{G} \to (-\infty, +\infty]$ proper lower-semicontinuous convex functions, and $L : \mathcal{H} \to \mathcal{G}$ bounded linear. Assume

$$Z = \{(x, y) : -L^*y \in \partial f(x),\ Lx \in \partial h(y)\} \neq \varnothing.$$

For the usual saddle problem, $h = g^*$. For $\tau, \sigma > 0$ and arbitrary x_0, y_0, use exactly

$$\begin{aligned} x_n &= \operatorname{prox}_{\tau f}(x_{n-1} - \tau L^* y_{n-1}), \\ y_n &= \operatorname{prox}_{\sigma h}\left(y_{n-1} + \frac{\sigma}{4} L(5x_n - x_{n-1})\right), \quad n \geq 1. \end{aligned} \tag{1}$$

This is a primal-first form of the Chambolle–Pock method [1]; the primal point is updated first and extrapolated with $\theta = 1/4$. The initial points need not lie in the effective domains. Positive-index proximal points do.

Theorem 1. *Put $\rho = \tau\sigma\|L\|^2$. If $\rho \leq 8/3$, every trajectory of (1) is bounded. If $\rho < 8/3$, then*

$$\sum_{n\geq 1} \left(\|x_n - x_{n-1}\|^2 + \|y_n - y_{n-1}\|^2\right) < \infty,$$

and the full trajectory converges weakly to a point of Z. In finite dimensions it converges in norm. The open threshold is sharp for a theorem applying to every problem in this class.

Theorem 2. *If $\rho = 8/3$, there are states $z_{\mathrm{e}}, z_{\mathrm{o}}$ such that*

$$(x_{2n}, y_{2n}) \rightharpoonup z_{\mathrm{e}}, \qquad (x_{2n+1}, y_{2n+1}) \rightharpoonup z_{\mathrm{o}}.$$

They form an exact period-two orbit of (1)*, allowing a constant orbit. Their midpoint belongs to Z. The adjacent midpoints and ordinary Cesàro averages converge weakly to that midpoint, and*

$$\sum_{n\geq 2} \big(\|x_n - x_{n-2}\|^2 + \|y_n - y_{n-2}\|^2\big) < \infty.$$

In finite dimensions all asserted convergences are in norm.

Two-step differences here mean $z_n - z_{n-2}$, not $z_n - 2z_{n-1} + z_{n-2}$. Neither first-difference regularity nor full point convergence holds in general at equality.

2 Normalization and the exact certificate

Choose $r > 0$ with $\rho \leq r \leq 8/3$, choosing $r < 8/3$ for an open-range conclusion. Set

$$X = \sqrt{r/\tau}\, x, \quad Y = y/\sqrt{\sigma}, \quad K = \sqrt{\tau\sigma/r}\, L.$$

$$F(X) = f(\sqrt{\tau/r}\, X), \qquad H(Y) = h(\sqrt{\sigma} Y).$$

Then $\|K\| \leq 1$ and the updates become

$$X_n = \mathrm{prox}_{rF}(X_{n-1} - rK^*Y_{n-1}), \qquad Y_n = \mathrm{prox}_H\left(Y_{n-1} + \tfrac{1}{4}K(5X_n - X_{n-1})\right). \quad (2)$$

For a normalized KKT point $p = (U, V)$, center $u = X - U, v = Y - V$ and use the nonnegative functions

$$F_p(u) = F(U + u) - F(U) + \langle K^*V, u\rangle, \quad H_p(v) = H(V + v) - H(V) - \langle KU, v\rangle.$$

They vanish at zero with zero subgradients. Actual proximal optimality gives

$$A_i = \frac{u_{i-1} - u_i}{r} - K^*v_{i-1} \in \partial F_p(u_i), \quad B_i = v_{i-1} - v_i + \tfrac{1}{4}K(5u_i - u_{i-1}) \in \partial H_p(v_i). \quad (3)$$

2.1 Formal coefficient identities

The following finite certificate is also specified numerically, with exact rational numbers, in Appendix A. Initially put $r_* = 8/3$. Order ten scalar slots as

$$z = (X_{-3}, Y_{-3}, X_{-2}, Y_{-2}, X_{-1}, Y_{-1}, X_0, Y_0, X_1, Y_1)^T$$

and order ten value slots $v = (F_{-3}, H_{-3}, \ldots, F_1, H_1)^T$ likewise. Let $e(i, t)$ select coordinate $2(i + 3) + t$ with zero-based numbering. At $i = -2, -1, 0, 1$ use formal gradients

$$A_i = \tfrac{3}{8}(X_{i-1} - X_i) - Y_{i-1}, \qquad B_i = Y_{i-1} - Y_i + \tfrac{1}{4}(5X_i - X_{i-1}).$$

The anchor $*$ has zero vector, value and gradient. Define slacks

$$s(i,j,0) = F_j - F_i - \langle A_i, X_j - X_i\rangle, \qquad s(i,j,1) = H_j - H_i - \langle B_i, Y_j - Y_i\rangle.$$

For source $*$ these are just target values; for target $*$ they retain the source gradient. Set

$$\begin{aligned} 571 S_0 = {} & 71 s(0,-2,0) + 63 s(0,-1,0) + 205 s(0,1,0) \\ & + 22 s(0,-3,1) + 48 s(0,-2,1) + 162 s(0,1,1). \end{aligned}$$

Let z_-, z_+ be the old and new four-pair histories, and v_-, v_+ their values. With $P = P^T \in \mathbb{R}^{8\times 8}$ and $c \in \mathbb{R}^8$, put

$$E_\pm = \langle z_\pm, P z_\pm\rangle + c^T v_\pm.$$

The appendix coefficients define nonnegative multipliers λ, ν and residual matrices D, C by the exact identities

$$E_- - E_+ = S_0 + \sum \lambda_{ijt} s(i,j,t) + \langle z, Dz\rangle, \tag{4}$$

$$E_- - \|X_0\|^2 - \|Y_0\|^2 = \sum \nu_{ijt} s(i,j,t) + \langle z, Cz\rangle. \tag{5}$$

Every value coefficient cancels identically in both equations. Signed entries of c are retained, not discarded.

Let Z stack five I_2 matrices and let

$$a = (-1, -3/4, 1, 3/4, -1, -3/4, 1, 3/4, -1, -3/4)^T.$$

Exact rational elimination gives

$$D \succeq 0, \quad \ker D = \operatorname{ran}[Z\ a], \quad C_{0:8,0:8} \succ 0, \quad C_{8:10,:} = C_{:,8:10} = 0. \tag{6}$$

For Z_4 stacking four I_2 matrices,

$$M = Z_4^T P Z_4 = \begin{pmatrix} \frac{10079520903}{1142000000} & -\frac{3796175883}{1142000000} \\ -\frac{3796175883}{1142000000} & \frac{3359840301}{142750000} \end{pmatrix} \succ 0. \tag{7}$$

Here is an explicit way to certify all signs rather than rely on numerical eigenvalues. The columns of

$$T = \begin{pmatrix} 3/4 & -1 & 0 & 0 & 3/4 & -1 & 0 \\ -1 & 0 & -1 & 0 & -1 & 0 & -1 \\ -3/4 & 0 & 0 & -1 & -3/4 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

span $(\mathrm{ran}[Z\ a])^{\perp}$. Exact arithmetic verifies $D[Z\ a] = 0$, $T^T[Z\ a] = 0$, and strictly positive pivots in the unpivoted LDL^T factorizations of $T^T DT$, $C_{0:8,0:8}$ and M. This proves (6)–(7). All inputs and the factorization algorithm are supplied in the reproducibility files. For a symmetric matrix Q, the algorithm uses

$$d_j = Q_{jj} - \sum_{k<j} L_{jk}^2 d_k, \quad L_{ij} = \frac{Q_{ij} - \sum_{k<j} L_{ik} L_{jk} d_k}{d_j}, \quad L_{jj} = 1.$$

Thus these are finite rational sign claims, independently reproducible without a solver.

2.2 The open-range family

Put $\alpha = (8/3)/r > 1$. Multiply only primal–primal entries of P by α, retaining c and all slack weights, and replace the primal coefficient $3/8$ by $3\alpha/8$. If Q_x retains the even–even principal block of Q and is zero elsewhere, and $E_x = e(0,0)e(0,0)^T$, exact coefficient comparison gives

$$\begin{aligned} D(r) &= D + (\alpha - 1)D_x, \\ C(r) &= C + (\alpha - 1)(C_x + E_x), \\ M(r) &= M + (\alpha - 1)\,\mathrm{diag}(M_{11}, 0). \end{aligned} \tag{8}$$

All value identities remain exact. The additional E_x occurs because the coercivity norm in (5) is not scaled. Principal-block positivity gives $D_x, C_x \succeq 0$. Direct rational calculation gives

$$D_x Z = 0, \qquad a^T D_x a = \frac{75242281851}{142750000} > 0.$$

Since every vector in $\ker D$ is $Zb + ta$, the intersection $\ker D \cap \ker D_x$ is $\mathrm{ran}\, Z$. The kernel of a sum of positive-semidefinite matrices with positive coefficients is the intersection of their kernels. Hence

$$\ker D(r) = \mathrm{ran}\, Z \quad (0 < r < 8/3).$$

Consequently $D(r) \succeq \delta(r) Q_{\mathrm{inc}}$ for some $\delta(r) > 0$, where

$$z^T Q_{\mathrm{inc}} z = |X_1 - X_0|^2 + |Y_1 - Y_0|^2.$$

For an explicit constant, let U_0 have columns $e_j - e_{j \bmod 2}$, $j = 2, \ldots, 9$. Set

$$\eta = \left\{ \mathrm{tr}[(U_0^T (D + D_x) U_0)^{-1} U_0^T U_0] \right\}^{-1} > 0.$$

The denominator is the sum of reciprocal eigenvalues of $D + D_x$ on $\mathrm{ran}\, Z^{\perp}$, so $D + D_x \succeq \eta \Pi_{Z^{\perp}}$. Since $\|Q_{\mathrm{inc}}\| = 2$ and it kills Z,

$$\delta(r) = \tfrac{1}{2}\eta \min\{1, (8/3)/r - 1\}$$

is valid. It tends to zero at the endpoint, as it must.

2.3 Transfer to the original Hilbert spaces

In $\mathcal{V} = \mathcal{G} \oplus \mathcal{H}$ define isometries

$$Ju = (Ku, (I - K^*K)^{1/2}u), \qquad Iv = (v, 0).$$

They preserve same-variable norms and satisfy $\langle Ju, Iv\rangle = \langle Ku, v\rangle$. Substitute Ju_i, Iv_i into the scalar certificate while retaining the actual centered function values. Every formal slack then equals the genuine convexity slack from (3). No embedded function or invented subgradient is needed. Scalar positive-semidefinite matrix inequalities tensor with the identity of $\mathcal{V}$: expand in an orthonormal basis of the finite-dimensional span of the finitely many vectors if desired. Apply the history identity at $n \geq 4$ so the oldest value index is at least one. Thus every value is finite.

3 Boundedness, convergence, and sharpness

The identities imply, for every KKT anchor p, a finite storage $E_n(p)$ with

$$E_n(p) \geq \|X_n - U\|^2 + \|Y_n - V\|^2, \qquad E_n(p) \geq E_{n+1}(p), \quad n \geq 4.$$

This proves boundedness. For $r < 8/3$, the increment margin and telescoping prove square summability of first differences. Fixed coordinate scalings preserve it.

The current source-anchor weights are

$$\lambda_{*,0,0} = \frac{3170910681}{285500000} > 0, \qquad \lambda_{*,0,1} = \frac{814925459}{114200000} > 0.$$

Thus $F_p(X_n - U)$ and $H_p(Y_n - V)$ are summable, also at the endpoint. Every signed finite combination of their recent values tends to zero. In the open range, vanishing finite-lag differences give

$$E_n(p) = \|z_n - p\|_{Q_r}^2 + o(1), \qquad Q_r = \begin{pmatrix} M(r)_{11}I & M(r)_{12}K^* \\ M(r)_{12}K & M(r)_{22}I \end{pmatrix},$$

where $z_n = (X_n, Y_n)$. The operator Q_r is bounded and strongly positive because $M(r) \succ 0$ and $\|K\| \leq 1$. It does not depend on p.

Define the maximal monotone operator

$$\mathcal{T}(X, Y) = (\partial F(X) + K^*Y, \partial H(Y) - KX).$$

The product subdifferential is maximal monotone; adding the bounded everywhere-defined skew operator preserves maximality. Actual optimality provides

$$w_n = (-\Delta X_n/r + K^*\Delta Y_n, -\Delta Y_n + \tfrac{1}{4}K\Delta X_n) \in \mathcal{T}z_n.$$

In the open range, $w_n \to 0$. If $z_{n_j} \rightharpoonup z$, monotonicity against any $(v, q) \in \operatorname{graph}\mathcal{T}$ yields $\langle z - v, -q\rangle \geq 0$, hence $0 \in \mathcal{T}z$. Every weak cluster point is KKT. For any two

KKT points p, q, both squared Q_r distances have limits. Their difference is affine weakly continuous. At two distinct cluster points p, q its subsequential values would be $-\|p - q\|_{Q_r}^2$ and $+\|p - q\|_{Q_r}^2$, impossible. Bounded Hilbert sequences admit weakly convergent subsequences; uniqueness of the weak cluster point proves weak convergence. In finite dimensions it is norm convergence.

For sharpness take $F = H = 0$, scalar $K = 1$. The update matrix is

$$R_r = \begin{pmatrix} 1 & -r \\ 1 & 1 - 5r/4 \end{pmatrix}, \qquad \det(I + R_r) = 4 - 3r/2.$$

At $r = 8/3$, $(1, 3/4)$ is an eigenvector with eigenvalue -1, giving the exact nonconstant orbit $\pm(1, 3/4)$. For $r > 8/3$, the characteristic polynomial is negative at -1 and positive for sufficiently negative arguments, so it has a real root below -1; the corresponding orbit is unbounded. This proves the sharp universal boundary and shows why its extra kernel direction cannot be removed.

4 Endpoint midpoint convergence

Now $r = 8/3$. Write $d_n = z_n - z_{n-1} = (a_n, b_n)$ and $m_n = (z_n + z_{n-1})/2$. The following linear maps on the five-pair lifted history annihilate $\ker D$:

$$J(X_{n+1} - X_{n-1}), \quad I(Y_{n+1} - Y_{n-1}), \quad J(X_{n+1} - X_n) - \tfrac{4}{3} I(Y_{n+1} - Y_n).$$

For a finite positive-semidefinite matrix D, if $\ker D \subset \ker B$, then $B^*B \preceq cD$ for a finite c, by restricting to $(\ker D)^\perp$. Tensoring and telescoping therefore give

$$\sum_n \|z_n - z_{n-2}\|^2 < \infty, \qquad \sum_n \|Ja_n - \tfrac{4}{3} Ib_n\|^2 < \infty. \tag{9}$$

In particular $d_n + d_{n-1} \to 0$ and $m_n - m_{n-1} \to 0$.

The graph elements above are $w_n = Rd_n$, where

$$R(a, b) = (-3a/8 + K^*b, -b + Ka/4).$$

Thus $w_n + w_{n-1} \to 0$. Moreover,

$$\langle d_n, w_n \rangle = -\tfrac{3}{8}\|a_n\|^2 - \|b_n\|^2 + \tfrac{5}{4}\langle Ka_n, b_n \rangle \longrightarrow 0.$$

Indeed bounded increments and (9) imply $\|a_n\|^2 - (16/9)\|b_n\|^2 \to 0$ and $\langle Ka_n, b_n \rangle - (4/3)\|b_n\|^2 \to 0$; the remaining coefficient is $-(3/8)(16/9) - 1 + (5/4)(4/3) = 0$.

For any $(v, q) \in \operatorname{graph} \mathcal{T}$, average the two actual monotonicity inequalities to obtain

$$0 \le \langle m_n - v, (w_n + w_{n-1})/2 - q \rangle + \tfrac{1}{4}\langle d_n, w_n - w_{n-1} \rangle.$$

The last term tends to zero since it equals $\frac{1}{2}\langle d_n, w_n\rangle - \frac{1}{4}\langle d_n, w_n + w_{n-1}\rangle$. Every weak midpoint cluster point is therefore a zero of $\mathcal{T}$, by maximality. This does not assert that averaged graph points lie in the graph.

To prove uniqueness, let $\mathcal{L}(X, Y) = (JX, IY)$ and let h_n stack the four vectors $\mathcal{L}z_{n-3}, \dots, \mathcal{L}z_n$. Summable shifted values give

$$E_n(p) = \langle h_n - Z_4\mathcal{L}p, P(h_n - Z_4\mathcal{L}p)\rangle + o(1).$$

For two KKT anchors subtract these expressions and average at $n, n-1$. Since

$$(h_n + h_{n-1})/2 = Z_4\mathcal{L}m_n + o(1),$$

we obtain

$$\frac{E_n(p) + E_{n-1}(p) - E_n(q) - E_{n-1}(q)}{2} = \|m_n - p\|_Q^2 - \|m_n - q\|_Q^2 + o(1),$$

with $Q = Q_{8/3} \succ 0$. The left side has a limit. The same affine weak-continuity argument used above makes the weak midpoint cluster point unique. Thus $m_n \rightharpoonup p \in Z$. Finally

$$\frac{1}{N}\sum_{n=1}^{N} z_n - \frac{1}{N}\sum_{n=1}^{N} m_n = (z_N - z_0)/(2N) \to 0,$$

so ordinary Cesàro means have the same weak limit.

5 Cycle classification and selection of its phase

Proposition 5.1. The exact period-two orbits of the normalized endpoint map are precisely

$$(U + a, V + b), \quad (U - a, V - b),$$

where (U, V) is KKT and

$$\begin{aligned} K^*Ka = a, \qquad b = \tfrac{3}{4}Ka, \\ -K^*V \in \partial F(U + a) \cap \partial F(U - a), \\ KU \in \partial H(V + b) \cap \partial H(V - b). \end{aligned} \tag{10}$$

Proof. The primal subgradients at the two states are $-K^*V \pm (K^*b - 3a/4)$, and the dual ones are $KU \pm (3Ka/2 - 2b)$. Their monotonicity gives

$$\langle Ka, b\rangle \geq \tfrac{3}{4}\|a\|^2, \qquad \langle Ka, b\rangle \geq \tfrac{4}{3}\|b\|^2.$$

Writing $A = \|a\|, B = \|b\|$, contraction and Cauchy–Schwarz give

$$AB \geq \langle Ka, b\rangle \geq \max\{3A^2/4, 4B^2/3\} \geq AB.$$

Equality forces $b = 3Ka/4$ and $K^*Ka = a$, including the zero case. The subgradients reduce to those in (10). A common supporting subgradient at two endpoints supports their midpoint; hence (U, V) is KKT. Conversely these inclusions are exactly the two proximal optimality conditions in both directions, so they generate the asserted orbit. ∎

Let $p = (U, V)$ be the midpoint limit, and write the nonnegative tilted functions in uncentered coordinates as

$$\phi(X) = F(X) - F(U) + \langle K^*V, X - U\rangle, \quad \psi(Y) = H(Y) - H(V) - \langle KU, Y - V\rangle.$$

They vanish along the iterates in the limit. Use actual uncentered subgradients A_n, B_n from (2). In addition to the source-anchor values, the certificate has strictly positive current iterate-to-anchor weights

$$\lambda_{0,*,0} = 140537/10000, \qquad \lambda_{0,*,1} = 2295593/200000.$$

Consequently the nonnegative supporting slacks

$$\begin{aligned} q_n &= \langle A_n + K^*V, X_n - U\rangle - \phi(X_n), \\ t_n &= \langle B_n - KU, Y_n - V\rangle - \psi(Y_n) \end{aligned}$$

are summable. These additional slacks will select the phase.

Take any weak cluster subsequence $z_{2n_j} \rightharpoonup (U + c, V + d)$. Midpoint convergence gives $z_{2n_j-1} \rightharpoonup (U - c, V - d)$. The strongly vanishing lifted increment relation gives $Jc = (4/3)Id$, so $K^*Kc = c$ and $d = 3Kc/4$. Weak lower semicontinuity of the nonnegative convex functions and their vanishing values give

$$\phi(U + c) = \phi(U - c) = \psi(V + d) = \psi(V - d) = 0.$$

By Proposition 5.1 these cluster endpoints are a genuine cycle, even in infinite-dimensional Hilbert spaces. This uses lower semicontinuity of values, not weak–weak graph closure.

Let C_p be the set of all amplitudes a satisfying (10), with $b = 3Ka/4$. For any fixed $a \in C_p$, the subgradient inequalities at X_n applied to $U \pm a$ give

$$|\langle A_n + K^*V, a\rangle| \le q_n.$$

Similarly $|\langle B_n - KU, b\rangle| \le t_n$. Define scalar sequences

$$u_n = \langle X_n - U, a\rangle, \quad v_n = \langle Y_n - V, Ka\rangle, \quad g_n = \langle A_n + K^*V, a\rangle, \quad k_n = \langle B_n - KU, Ka\rangle.$$

Both g_n and k_n are absolutely summable. Projecting actual optimality gives

$$g_n = -\tfrac{3}{8}(u_n - u_{n-1}) - v_{n-1}, \qquad k_n = -(v_n - v_{n-1}) + \tfrac{1}{4}(5u_n - u_{n-1}).$$

Thus $s_n = u_n - 2v_n$ satisfies

$$s_n = -s_{n-1} + 4g_n + 2k_n.$$

It follows that $(-1)^n s_n$ converges. For every $a \in C_p$, the even scalar expression $u_{2n} - 2v_{2n}$ therefore has a limit.

If $c, d \in C_p$ are two even cluster amplitudes, these scalar limits equal $-\frac{1}{2}\langle c, a\rangle$ and $-\frac{1}{2}\langle d, a\rangle$ respectively, because $K^*Ka = a$. Hence $\langle c-d, a\rangle = 0$ for every $a \in C_p$. Choosing $a = c$ and $a = d$ gives $\|c-d\|^2 = 0$. There is only one even weak cluster point, so the even subsequence converges weakly. The relation $z_{2n+1} = 2m_{2n+1} - z_{2n}$ proves convergence of the odd subsequence. This proves Theorem 2.

Corollary 5.1. *Every endpoint trajectory converges weakly to a KKT point if and only if no KKT midpoint admits a nonzero amplitude satisfying* (10)*. In particular, strict convexity of either f or h on its effective domain suffices. In finite dimensions convergence is in norm.*

Proof. If there are no nonzero amplitudes, the selected endpoints coincide. Conversely any nonzero admissible amplitude produces an exact nonconstant cycle, whose trajectory cannot converge weakly to one point. A strictly convex function cannot have the same affine support at two distinct points. If $a \neq 0$ and $K^*Ka = a$, then also $b = 3Ka/4 \neq 0$. Thus strict convexity of either function excludes the cycle. ∎

6 Relation to earlier convergence results

The primal-first algorithm is the same recurrence considered by Banert, Upadhyaya and Giselsson [2] and by Upadhyaya [3]. Their general formula $4/(1+2\theta)$ and its bilinear necessity are prior results; neither that necessary bound nor the scalar endpoint cycle is claimed here as new. The main published sharp sufficiency theorem in [2] assumes $\theta > 1/2$. Its proof uses a convexity weight $\theta - 1/2$, so substitution of $\theta = 1/4$ into that proof is invalid, rather than a counterexample to its stated theorem. Upadhyaya's version 2 theorem [3] treats small extrapolation with

$$\tau\sigma\|L\|^2 < \frac{4\theta(2-\theta)}{1-2\theta+9\theta^2-4\theta^3};$$

at $\theta = 1/4$ the right side is exactly $7/4$. The present certificate reaches the known necessary boundary and resolves its endpoint through phase selection.

Longer-history Lyapunov searches and convex interpolation have important methodological precedents, including AutoLyap [4]. Numerical feasibility at sampled parameters is distinct from the exact symbolic family and Hilbert-space transfer proved here. Earlier smaller quarter-extrapolation bounds are subsumed by Theorem 1, not presented as separate advances.

Outer relaxation changes the map: Algorithm 4 in [5] requires its relaxation $\varrho < \min\{2, 2\theta\}$, so the unrelaxed choice $\varrho = 1$ excludes $\theta = 1/4$. The prediction–correction specialization of Bai–Rao [6] likewise contains a correction and relaxation; setting its

prediction parameter to 1/4 alone does not recover (1). These are scope distinctions, not claims that the cited theorems are false. An additional coupled-extrapolation paper [7] was screened only at abstract level; a full theorem-by-theorem exclusion of all its specializations is not asserted. The positioning is therefore source-specific and does not constitute an exhaustive priority guarantee. No conclusion is drawn here for $\theta = 1/8$ or for all small extrapolation parameters.

A Complete rational certificate

The ordered coefficient vector consists of the row-major upper triangle of P (36 entries), followed by c (8 entries), the descent weights (48 entries), and the coercivity weights (32 entries). Tables below specify all coefficients. The symbol $*$ denotes the zero anchor. Together with (4)–(5), these data define D and C uniquely. The accompanying relative directory `reproducibility/` contains `exact_certificate.json`, the standalone `verify_certificate.py`, and `exact_results.json`. Run the verifier with Python and SymPy from any directory; it uses its embedded rational data and writes the result beside itself. No numerical optimization package or discovery output is required. The checks include all coefficient cancellations, ranks, positive rational pivots, the symbolic scaling identities, and the sharp scalar obstruction.

A.1 Quadratic and value coefficients

i	j	P_{ij}
1	1	3063926659/2284000000
1	2	−299389601/114200000
1	3	111375322693/9136000000
1	4	10227294081/2284000000
1	5	−65248258673/4568000000
1	6	3727465193/228400000
1	7	21728478149/9136000000
1	8	5078357393/2284000000
2	2	1705225409/285500000
2	3	−24148815423/571000000
2	4	582231/25000
2	5	15609666709/913600000
2	6	−42556839/2000000
2	7	−29974786641/4568000000
2	8	−6702477/2000000
3	3	−19442656801/1142000000
3	4	51918895643/2284000000
3	5	32593073587/4568000000
3	6	13746666129/1142000000
3	7	−239142731/456800000

i	j	P_{ij}
3	8	−1826798139/2284000000
4	4	−10638353/1000000
4	5	−10227753/1000000
4	6	−13129833/1000000
4	7	−4200709/200000
4	8	5215597/500000
5	5	33078757/1000000
5	6	−26951307/1000000
5	7	−4659001/200000
5	8	767029/31250
6	6	71680741/1000000
6	7	13757479/500000
6	8	−15766753/500000
7	7	6056791/250000
7	8	−19801889/1000000
8	8	3458313/125000

i	c_i
1	10945493/200000
2	69700939/1000000
3	−317827/1000000
4	10017357/1000000
5	11895273/1000000
6	−27433319/500000
7	−4793433/125000
8	−17086073/1000000

A.2 Descent multipliers

Index	i	j	t	Weight
1	$*$	−3	0	11612243/1000000
2	$*$	−2	0	5141623471/571000000
3	$*$	−1	0	4953214009/571000000
4	$*$	0	0	3170910681/285500000
5	$*$	1	0	3811472813/571000000
6	−2	$*$	0	1238582511/142750000
7	−1	$*$	0	9478903/1000000
8	0	$*$	0	140537/10000
9	1	$*$	0	14863881/1000000
10	$*$	−3	1	5480426083/571000000
11	$*$	−2	1	6013220967/571000000
12	$*$	−1	1	3011079/200000
13	$*$	0	1	814925459/114200000
14	$*$	1	1	1356908759/285500000

Index	i	j	t	Weight
15	-2	$*$	1	132253607/14275000
16	-1	$*$	1	1244031/62500
17	0	$*$	1	2295593/200000
18	1	$*$	1	6425907/1000000
19	-2	-3	0	0
20	-2	-1	0	57100001577/571000000
21	-2	0	0	0
22	-2	1	0	0
23	-1	-3	0	28321319/1000000
24	-1	-2	0	0
25	-1	0	0	100
26	-1	1	0	0
27	0	-3	0	14793903/1000000
28	0	-2	0	7976989/250000
29	0	-1	0	0
30	0	1	0	100
31	1	-2	0	6297201/500000
32	1	-1	0	10307089/250000
33	1	0	0	0
34	-2	-3	1	0
35	-2	-1	1	84937779/1000000
36	-2	0	1	9862561/500000
37	-2	1	1	366831/1000000
38	-1	-3	1	12453037/250000
39	-1	-2	1	0
40	-1	0	1	44670047/500000
41	-1	1	1	692919/31250
42	0	-3	1	5126159/500000
43	0	-2	1	2336649/62500
44	0	-1	1	0
45	0	1	1	18897629/1000000
46	1	-2	1	6609369/1000000
47	1	-1	1	16352977/1000000
48	1	0	1	0

A.3 Coercivity multipliers

Index	i	j	t	Weight
1	$*$	-3	0	14339067/1000000
2	$*$	-2	0	3975341/125000
3	$*$	-1	0	21950509/500000
4	$*$	0	0	37313583/1000000
5	$*$	-3	1	3693087/250000
6	$*$	-2	1	4454109/125000
7	$*$	-1	1	492892/15625

Index	i	j	t	Weight
8	$*$	0	1	11680489/500000
9	-2	$*$	0	19780117/1000000
10	-1	$*$	0	16853889/500000
11	0	$*$	0	22955527/500000
12	-2	$*$	1	6147469/250000
13	-1	$*$	1	2062303/50000
14	0	$*$	1	6341953/200000
15	-2	-3	0	2229517/200000
16	-2	-1	0	350669/8000
17	-2	0	0	24668907/1000000
18	-1	-3	0	8736347/500000
19	-1	-2	0	5927483/250000
20	-1	0	0	37078137/1000000
21	0	-3	0	11768119/1000000
22	0	-2	0	43599747/1000000
23	0	-1	0	36129171/1000000
24	-2	-3	1	4967511/250000
25	-2	-1	1	10430727/500000
26	-2	0	1	20240957/1000000
27	-1	-3	1	27240929/1000000
28	-1	-2	1	404397/20000
29	-1	0	1	6302823/200000
30	0	-3	1	3908809/500000
31	0	-2	1	19863483/500000
32	0	-1	1	6473887/500000

References

[1] A. Chambolle and T. Pock, A first-order primal-dual algorithm for convex problems with applications to imaging, *Journal of Mathematical Imaging and Vision* **40** (2011), 120–145. doi:10.1007/s10851-010-0251-1.

[2] S. Banert, M. Upadhyaya and P. Giselsson, The Chambolle–Pock method converges weakly with $\theta > 1/2$ and $\tau\sigma\|L\|^2 < 4/(1+2\theta)$, *Optimization Letters* **20** (2026), 503–520. doi:10.1007/s11590-025-02250-0.

[3] M. Upadhyaya, The Chambolle–Pock method converges weakly with $0 < \theta \leq 1/2$ and $\tau\sigma\|L\|^2 < 4\theta(2-\theta)/(1-2\theta+9\theta^2-4\theta^3)$, arXiv:2604.06423v2 (20 August 2026). `https://arxiv.org/abs/2604.06423v2`.

[4] M. Upadhyaya, S. Das Gupta, A. B. Taylor, S. Banert and P. Giselsson, *The AutoLyap software suite for computer-assisted Lyapunov analyses of first-order methods*, arXiv:2506.24076v2 (2 March 2026). `https://arxiv.org/abs/2506.24076v2`.

[5] M. Nilsson, A. Åkerman and P. Giselsson, *Extending Douglas–Rachford splitting for convex optimization*, arXiv:2511.19637v2 (11 December 2025). `https://arxiv.org/abs/2511.19637v2`.

[6] J. Bai and S. Rao, *A new insight on the prediction-correction framework with applications to first-order methods*, author-posted manuscript, updated 19 June 2026; journal record doi:10.1016/j.orl.2026.107487. Author-posted manuscript.

[7] J. Wu and F. Ma, A primal-dual algorithm with coupled extrapolation: bridging the Chambolle–Pock and Peaceman–Rachford methods, *Numerical Algorithms* **102** (2026), 2485–2523, doi:10.1007/s11075-025-02207-5. Abstract-level comparison only. `https://doi.org/10.1007/s11075-025-02207-5`.